

tent_count_2024

```
library(dplyr)
library(ggplot2)
library(plotrix)
library(scales)
library(janitor)
library(lme4)
library(car)
library(emmeans)
library(ggh4x)
library(performance)
library(DHARMA)
library(multcomp)
library(multcompView)
library(broom)
library(tibble)

#graphics.off()

setwd("~/Documents/GitHub/heating_lacerates_final")

select <- dplyr::select
filter <- dplyr::filter
mutate <- dplyr::mutate
lag <- dplyr::lag

## Read and clean 2024 data

ursa2024_data <- read.csv("~/Documents/GitHub/heating_lacerates_final/data/2024_tentcount.csv") %>%
  clean_names() %>%
  mutate(
    tent_count = suppressWarnings(as.numeric(tent_count)),
    id = factor(id),
    plate = factor(plate),
    well = factor(well),
    line = factor(line),
    temp = factor(temp),
    treatment = as.character(treatment),
    symbiosis = factor(symbiosis),
    lacerate = factor(lacerate),
    day = as.numeric(day),
    day_cat = as.character(day_cat)
  ) %>%
  mutate(
    treatment = gsub("APO", "Apo", treatment),
    treatment = gsub("SYM", "Sym", treatment),
    treatment = gsub("INO", "Ino", treatment),
```

```

treatment = gsub("25C$", "25", treatment),
treatment = gsub("32C$", "32", treatment),
day_cat = gsub("^[0-9]+_day$", "day_\\1", day_cat)
) %>%
mutate(
  treatment = factor(treatment),
  day_cat = factor(day_cat)
)

str(ursa2024_data)

## 'data.frame': 738 obs. of 11 variables:
## $ id : Factor w/ 82 levels "H2-Apo-25-A1-1",...: 1 1 1 1 1 1 1 1 1 2 ...
## $ plate : Factor w/ 6 levels "1","2","3","4",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ treatment : Factor w/ 6 levels "H2-Apo-25","H2-Apo-32",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ line : Factor w/ 1 level "H2": 1 1 1 1 1 1 1 1 1 1 ...
## $ symbiosis : Factor w/ 3 levels "Apo","Ino","Sym": 1 1 1 1 1 1 1 1 1 1 ...
## $ well : Factor w/ 12 levels "A1","A2","A3",...: 1 1 1 1 1 1 1 1 1 2 ...
## $ temp : Factor w/ 2 levels "25C (ambient)",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ lacerate : Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
## $ day : num 0 3 4 5 7 10 11 14 21 0 ...
## $ day_cat : Factor w/ 9 levels "day_0","day_10",...: 1 6 7 8 9 2 3 4 5 1 ...
## $ tent_count: num 0 NA NA NA NA NA NA NA NA 0 ...

table(ursa2024_data$treatment)

##
## H2-Apo-25 H2-Apo-32 H2-Ino-25 H2-Ino-32 H2-Sym-25 H2-Sym-32
## 117 108 117 108 144 144

table(ursa2024_data$day_cat)

##
## day_0 day_10 day_11 day_14 day_21 day_3 day_4 day_5 day_7
## 82 82 82 82 82 82 82 82 82

## Mortality plot
df_mortality <- ursa2024_data %>%
  filter(
    line == "H2",
    day_cat %in% c("day_14", "day_21"),
    symbiosis != "Inoc",
    !treatment %in% c("H2-Ino-25", "H2-Ino-32")
  ) %>%
  mutate(
    Mortality = ifelse(is.na(tent_count) | tent_count == 0, "Dead", "Alive"),
    dead = ifelse(Mortality == "Dead", 1, 0),
    Mortality = factor(Mortality, levels = c("Alive", "Dead")),
    day_cat = factor(day_cat, levels = c("day_14", "day_21"))
  )

nrow(df_mortality)

```

```
## [1] 114
```

```
table(df_mortality$day_cat)
```

```
##  
## day_14 day_21  
##      57      57
```

```
table(df_mortality$treatment)
```

```
##  
## H2-Apo-25 H2-Apo-32 H2-Ino-25 H2-Ino-32 H2-Sym-25 H2-Sym-32  
##          26          24           0           0          32          32
```

```
table(df_mortality$Mortality)
```

```
##  
## Alive  Dead  
##      51     63
```

```
my_theme <- theme_classic(base_size = 14) +  
  theme(  
    text = element_text(family = "sans", size = 14),  
    axis.text = element_text(size = 14, colour = "black"),  
    axis.title = element_text(size = 14),  
    legend.text = element_text(size = 14),  
    legend.title = element_text(size = 14)  
  )
```

```
mortality_plot_df_flipped <- df_mortality %>%  
  mutate(  
    sym_state = case_when(  
      grepl("Apo", treatment) ~ "Aposymbiotic",  
      grepl("Sym", treatment) ~ "Symbiotic"  
    ),  
    temp_label = case_when(  
      grepl("25", treatment) ~ "25°C",  
      grepl("32", treatment) ~ "32°C"  
    ),  
    day_label = case_when(  
      day_cat == "day_14" ~ "14 dpl",  
      day_cat == "day_21" ~ "21 dpl"  
    )  
  ) %>%  
  group_by(sym_state, temp_label, day_label) %>%  
  summarise(  
    mortality_prop = mean(dead, na.rm = TRUE),  
    .groups = "drop"  
  ) %>%  
  mutate(  
    survival_prop = 1 - mortality_prop,
```

```

temp_label = factor(temp_label, levels = c("25°C", "32°C")),
sym_state = factor(sym_state, levels = c("Aposymbiotic", "Symbiotic")),
outline_group = case_when(
  sym_state == "Aposymbiotic" & temp_label == "25°C" ~ "apo_25",
  sym_state == "Symbiotic" & temp_label == "25°C" ~ "sym_25",
  sym_state == "Aposymbiotic" & temp_label == "32°C" ~ "apo_32",
  sym_state == "Symbiotic" & temp_label == "32°C" ~ "sym_32"
),
group_order = case_when(
  sym_state == "Aposymbiotic" & day_label == "14 dpl" ~ 1,
  sym_state == "Aposymbiotic" & day_label == "21 dpl" ~ 2,
  sym_state == "Symbiotic" & day_label == "14 dpl" ~ 3,
  sym_state == "Symbiotic" & day_label == "21 dpl" ~ 4
),
axis_text = case_when(
  day_label == "14 dpl" ~ "14 dpl",
  day_label == "21 dpl" ~ "21 dpl"
)
) %>%
arrange(temp_label, group_order) %>%
group_by(temp_label, sym_state) %>%
mutate(
  bar_label = factor(axis_text, levels = rev(unique(axis_text)))
) %>%
ungroup()

survival_fill_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

outline_colors <- c(
  "sym_25" = "#3B6FB6",
  "apo_25" = "#6FA3D9",
  "sym_32" = "#E64B35",
  "apo_32" = "#F39B7F"
)

p_survival_base <- ggplot(
  mortality_plot_df_flipped,
  aes(x = bar_label)
) +
  coord_flip() +
  ggh4x::facet_nested(
    rows = vars(temp_label, sym_state),
    scales = "free_y",
    space = "free_y",
    switch = "y"
) +
  scale_y_continuous(
    labels = percent,

```

```

    limits = c(0, 1),
    expand = expansion(mult = c(0, 0.04))
  ) +
  labs(
    x = "Treatment Group",
    y = "Percent Survival"
  ) +
  my_theme +
  theme(
    panel.grid = element_blank(),
    strip.placement = "outside",
    strip.text.y.right = element_text(size = 14, face = "bold"),
    strip.background.y = element_rect(
      fill = "white",
      color = "black",
      linewidth = 1
    ),
    ggh4x.facet.nestline = element_blank(),
    panel.spacing.y = unit(0.08, "lines"),
    axis.text.y = element_text(size = 14, lineheight = 0.9),
    axis.text.x = element_text(size = 14),
    plot.margin = margin(8, 14, 8, 8)
  )

p_survival_flipped_outline <- p_survival_base +
  geom_col(
    aes(y = survival_prop),
    fill = "grey80",
    width = 0.82
  ) +
  geom_col(
    aes(y = 1, color = outline_group),
    fill = NA,
    linewidth = 0.9,
    width = 0.82
  ) +
  scale_color_manual(
    values = outline_colors,
    guide = "none"
  )

p_survival_flipped_colorbar <- p_survival_base +
  geom_col(
    aes(y = survival_prop, fill = outline_group),
    width = 0.82,
    color = NA
  ) +
  geom_col(
    aes(y = 1),
    width = 0.82,
    fill = NA,
    color = "black",
    linewidth = 0.9
  )

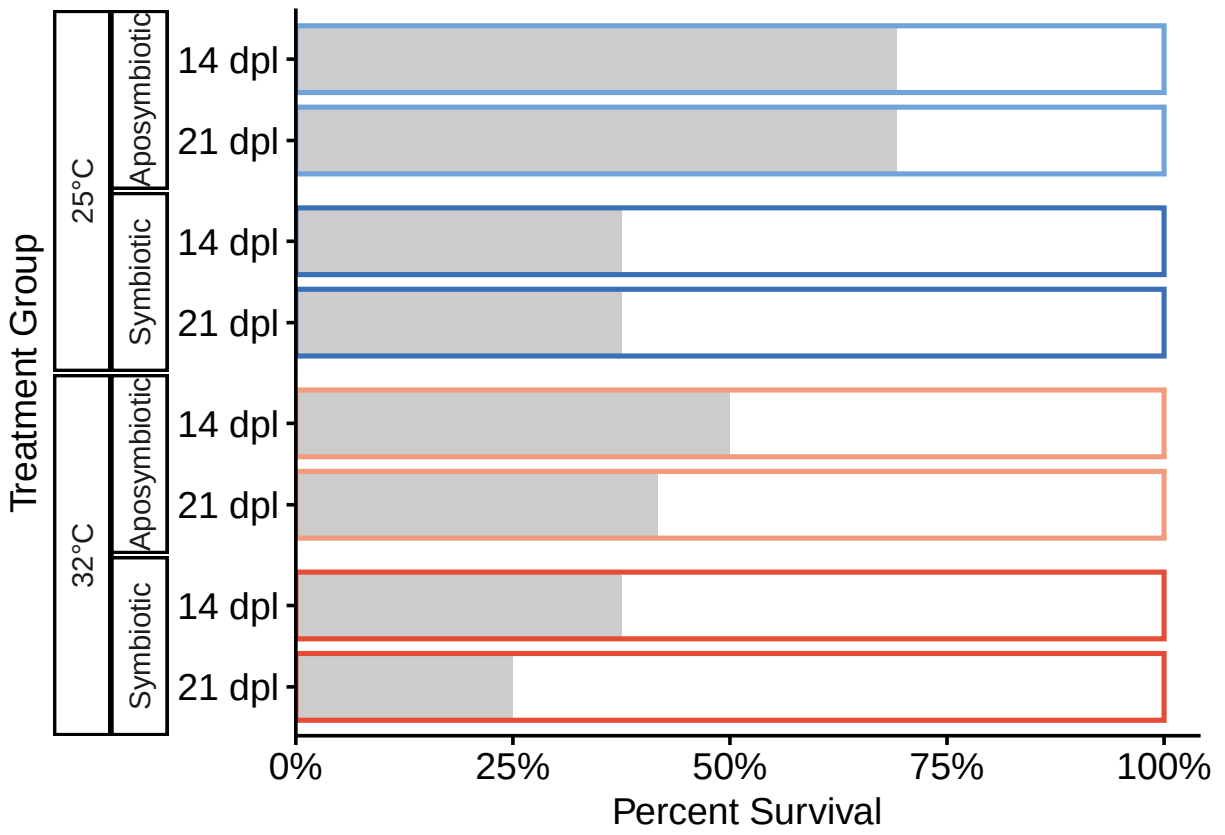
```

```

) +
  scale_fill_manual(
    values = survival_fill_colors,
    guide = "none"
  )

p_survival_flipped_outline

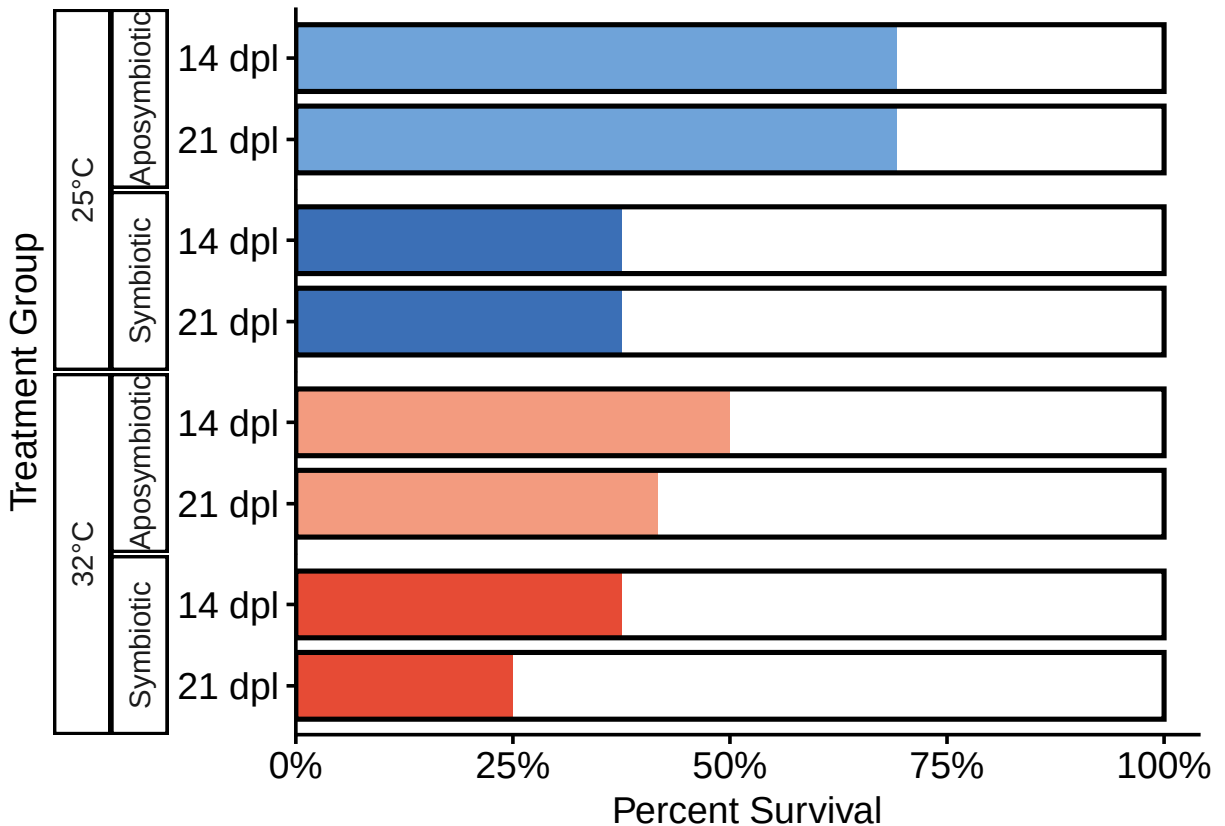
```



```

p_survival_flipped_colorbar

```



Mortality stats

```
mortality_glm <- glm(
  dead ~ temp * symbiosis,
  data = df_mortality,
  family = binomial
)
```

```
summary(mortality_glm)
```

```
##
## Call:
## glm(formula = dead ~ temp * symbiosis, family = binomial, data = df_mortality)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -0.8109    0.4249  -1.908  0.0563 .
## temp32C (heat stress)  0.9780    0.5902   1.657  0.0975 .
## symbiosisSym      1.3218    0.5603   2.359  0.0183 *
## temp32C (heat stress):symbiosisSym -0.7004    0.7919  -0.884  0.3765
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 156.77  on 113  degrees of freedom
```

```
## Residual deviance: 147.29 on 110 degrees of freedom
## AIC: 155.29
##
## Number of Fisher Scoring iterations: 4
```

```
car::Anova(mortality_glm, type = 3)
```

```
## Analysis of Deviance Table (Type III tests)
##
## Response: dead
##           LR Chisq Df Pr(>Chisq)
## temp           2.8284 1  0.09261 .
## symbiosis       5.8995 1  0.01515 *
## temp:symbiosis  0.7861 1  0.37528
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emm_mort_temp <- emmeans(mortality_glm, ~ temp | symbiosis)
emm_mort_sym <- emmeans(mortality_glm, ~ symbiosis | temp)
emm_mort_temp
```

```
## symbiosis = Apo:
## temp           emmean      SE df asymp.LCL asymp.UCL
## 25C (ambient)   -0.811 0.425 Inf   -1.644    0.0219
## 32C (heat stress) 0.167 0.410 Inf   -0.636    0.9700
##
## symbiosis = Sym:
## temp           emmean      SE df asymp.LCL asymp.UCL
## 25C (ambient)    0.511 0.365 Inf   -0.205    1.2265
## 32C (heat stress) 0.788 0.381 Inf    0.041    1.5360
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
emm_mort_sym
```

```
## temp = 25C (ambient):
## symbiosis emmean      SE df asymp.LCL asymp.UCL
## Apo       -0.811 0.425 Inf   -1.644    0.0219
## Sym        0.511 0.365 Inf   -0.205    1.2265
##
## temp = 32C (heat stress):
## symbiosis emmean      SE df asymp.LCL asymp.UCL
## Apo        0.167 0.410 Inf   -0.636    0.9700
## Sym        0.788 0.381 Inf    0.041    1.5360
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
## Survival stats
```

```

df_mortality <- df_mortality %>%
  mutate(
    alive = 1 - dead
  )

survival_glm <- glm(
  alive ~ temp * symbiosis,
  data = df_mortality,
  family = binomial
)

summary(survival_glm)

##
## Call:
## glm(formula = alive ~ temp * symbiosis, family = binomial, data = df_mortality)
##
## Coefficients:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)         0.8109     0.4249   1.908  0.0563 .
## temp32C (heat stress) -0.9780     0.5902  -1.657  0.0975 .
## symbiosisSym         -1.3218     0.5603  -2.359  0.0183 *
## temp32C (heat stress):symbiosisSym  0.7004     0.7919   0.884  0.3765
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 156.77  on 113  degrees of freedom
## Residual deviance: 147.29  on 110  degrees of freedom
## AIC: 155.29
##
## Number of Fisher Scoring iterations: 4

```

```

car::Anova(survival_glm, type = 3)

```

```

## Analysis of Deviance Table (Type III tests)
##
## Response: alive
##              LR Chisq Df Pr(>Chisq)
## temp          2.8284  1  0.09261 .
## symbiosis      5.8995  1  0.01515 *
## temp:symbiosis  0.7861  1  0.37528
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```

emm_surv_temp <- emmeans(survival_glm, ~ temp | symbiosis)
emm_surv_sym  <- emmeans(survival_glm, ~ symbiosis | temp)

emm_surv_temp

```

```

## symbiosis = Apo:

```

```
## temp          emmean    SE  df asymp.LCL asymp.UCL
## 25C (ambient)    0.811 0.425 Inf   -0.0219    1.644
## 32C (heat stress) -0.167 0.410 Inf   -0.9700    0.636
##
## symbiosis = Sym:
## temp          emmean    SE  df asymp.LCL asymp.UCL
## 25C (ambient)   -0.511 0.365 Inf   -1.2265    0.205
## 32C (heat stress) -0.788 0.381 Inf   -1.5360   -0.041
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
emm_surv_sym
```

```
## temp = 25C (ambient):
## symbiosis emmean    SE  df asymp.LCL asymp.UCL
## Apo       0.811 0.425 Inf   -0.0219    1.644
## Sym       -0.511 0.365 Inf   -1.2265    0.205
##
## temp = 32C (heat stress):
## symbiosis emmean    SE  df asymp.LCL asymp.UCL
## Apo      -0.167 0.410 Inf   -0.9700    0.636
## Sym      -0.788 0.381 Inf   -1.5360   -0.041
##
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
```

```
## Symbiotic state plot
```

```
data_means <- ursa2024_data %>%
  filter(line == "H2") %>%
  group_by(treatment, day) %>%
  summarise(
    mean = mean(tent_count, na.rm = TRUE),
    se = std.error(tent_count, na.rm = TRUE),
    .groups = "drop"
  )

my_theme <- theme_classic(base_size = 14) +
  theme(
    text = element_text(family = "sans", size = 14),
    axis.text = element_text(size = 14, colour = "black"),
    axis.title = element_text(size = 14),
    legend.text = element_text(size = 14),
    legend.title = element_text(size = 14)
  )

p_sym_state_shape <- ggplot(
  data = data_means %>%
    filter(treatment %in% c("H2-Apo-25", "H2-Ino-25", "H2-Sym-25")),
  aes(x = day, y = mean, group = treatment,
      shape = treatment, color = treatment)
) +
```

```

my_theme +

geom_line(color = "#3B6FB6", linewidth = 0.9) +

geom_errorbar(
  aes(ymin = mean - se, ymax = mean + se),
  width = 0.2,
  linewidth = 0.6
) +

geom_point(size = 4) +

annotate("text", x = 7, y = 10.2, label = "***", size = 6) +
annotate("text", x = 10, y = 12.0, label = "***", size = 6) +
annotate("text", x = 11, y = 12.0, label = "**", size = 6) +

ylab("Mean tentacle number") +
xlab("Days post laceration (dpl)") +

scale_y_continuous(breaks = seq(0, 13, 2), limits = c(0, 13)) +
scale_x_continuous(breaks = seq(min(data_means$day), max(data_means$day), 1)) +

scale_color_manual(
  values = c(
    "H2-Apo-25" = "grey50",
    "H2-Ino-25" = "#4DAF4A",
    "H2-Sym-25" = "#8C564B"
  ),
  labels = c(
    "H2-Apo-25" = "Apo 25°C",
    "H2-Ino-25" = "Inoc 25°C",
    "H2-Sym-25" = "Sym 25°C"
  )
) +

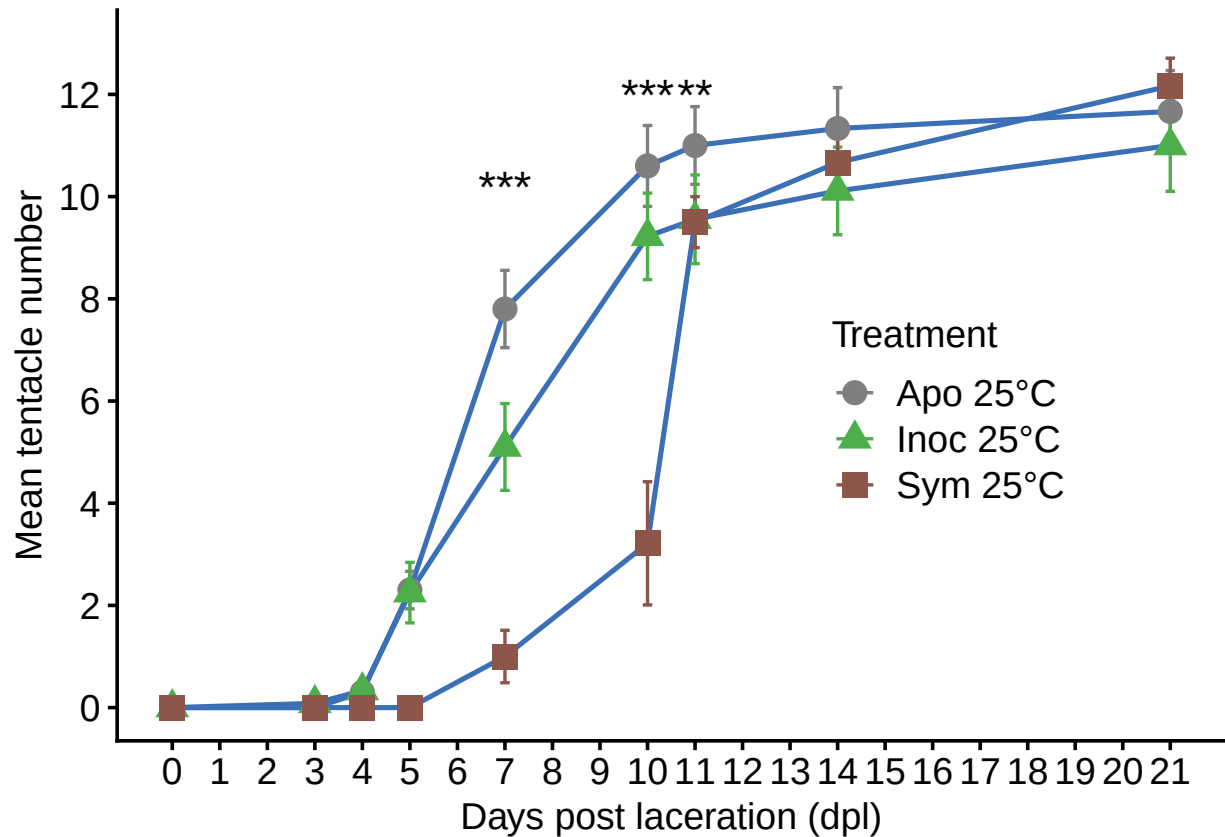
scale_shape_manual(
  values = c(
    "H2-Apo-25" = 16,
    "H2-Ino-25" = 17,
    "H2-Sym-25" = 15
  ),
  labels = c(
    "H2-Apo-25" = "Apo 25°C",
    "H2-Ino-25" = "Inoc 25°C",
    "H2-Sym-25" = "Sym 25°C"
  )
) +

labs(color = "Treatment", shape = "Treatment") +

theme(
  legend.position = c(0.75, 0.45),
  legend.justification = c("center", "center")
)

```

p_sym_state_shape



```
p_sym_state <- ggplot(
  data = data_means %>%
    filter(treatment %in% c("H2-Apo-25", "H2-Ino-25", "H2-Sym-25")),
  aes(x = day, y = mean, color = treatment, group = treatment)
) +
  my_theme +
  geom_line(linewidth = 0.9) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6
  ) +
  geom_point(size = 4) +

  annotate("text", x = 7, y = 10.2, label = "***", size = 6) +
  annotate("text", x = 10, y = 12.0, label = "***", size = 6) +
  annotate("text", x = 11, y = 12.0, label = "**", size = 6) +

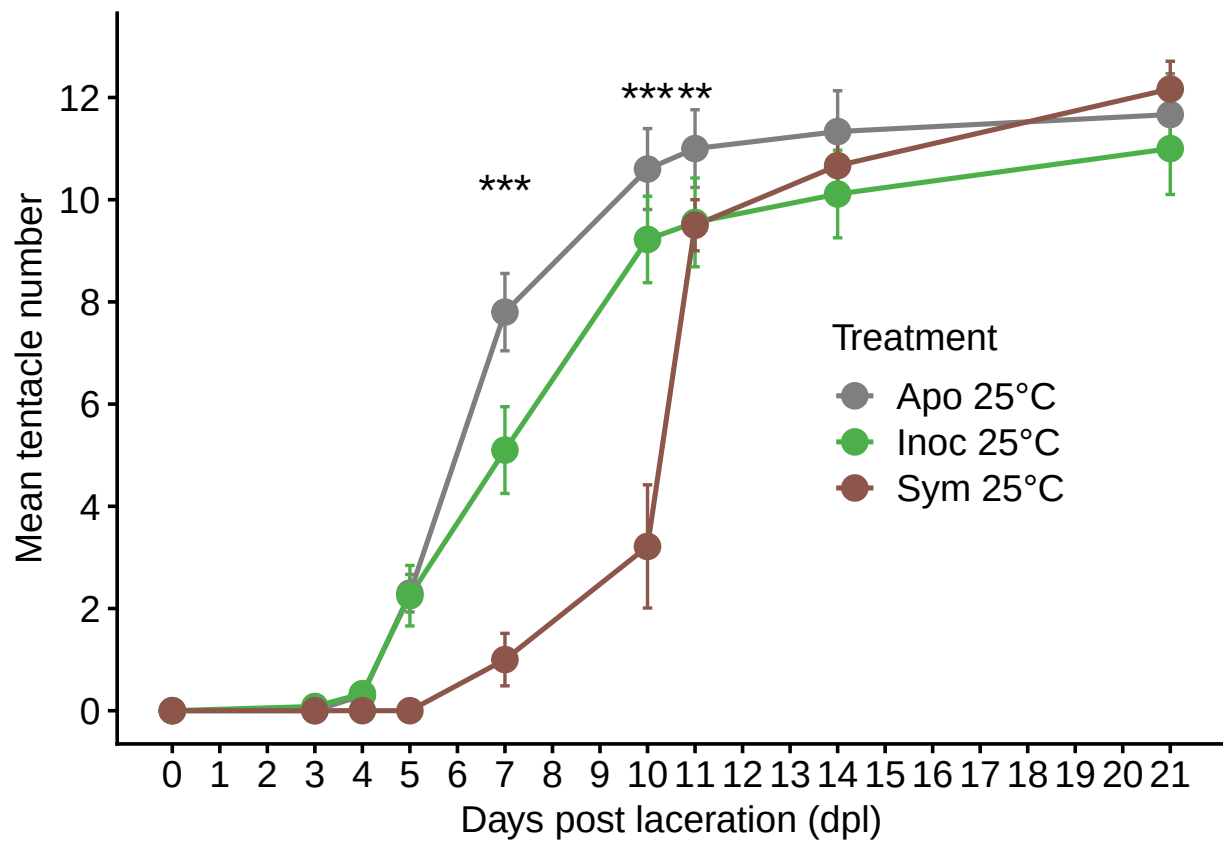
  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +
  scale_y_continuous(breaks = seq(0, 13, 2), limits = c(0, 13)) +
```

```

scale_x_continuous(breaks = seq(min(data_means$day), max(data_means$day), 1)) +
scale_color_manual(
  values = c(
    "H2-Apo-25" = "grey50",
    "H2-Ino-25" = "#4DAF4A",
    "H2-Sym-25" = "#8C564B"
  ),
  labels = c(
    "H2-Apo-25" = "Apo 25°C",
    "H2-Ino-25" = "Inoc 25°C",
    "H2-Sym-25" = "Sym 25°C"
  )
) +
labs(colour = "Treatment") +
theme(
  legend.position = c(0.75, 0.45),
  legend.justification = c("center", "center")
)

```

p_sym_state



```
## Inoc + Sym + Apo plot
```

```

p_sym_state_color_shape_2024 <- ggplot(
  data = data_means %>%

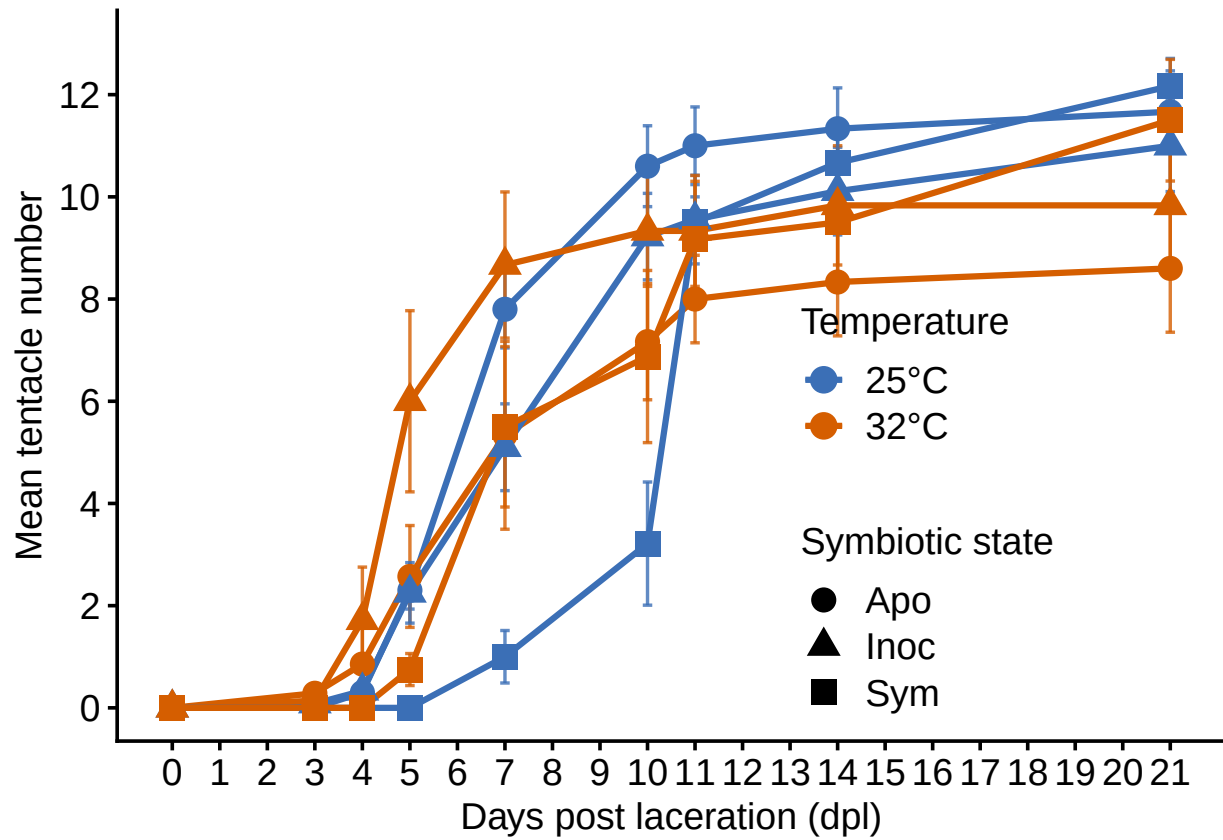
```

```

filter(treatment %in% c(
  "H2-Apo-25", "H2-Ino-25", "H2-Sym-25",
  "H2-Apo-32", "H2-Ino-32", "H2-Sym-32"
)) %>%
mutate(
  temp = ifelse(grepl("25", treatment), "25°C", "32°C"),
  state = case_when(
    grepl("Apo", treatment) ~ "Apo",
    grepl("Ino", treatment) ~ "Inoc",
    grepl("Sym", treatment) ~ "Sym"
  )
),
aes(
  x = day,
  y = mean,
  group = treatment,
  color = temp,
  shape = state
)
) +
my_theme +
geom_line(linewidth = 1.1) +
geom_errorbar(
  aes(ymin = mean - se, ymax = mean + se),
  width = 0.2,
  linewidth = 0.6,
  alpha = 0.8
) +
geom_point(size = 4) +
ylab("Mean tentacle number") +
xlab("Days post laceration (dpl)") +
scale_y_continuous(breaks = seq(0, 13, 2), limits = c(0, 13)) +
scale_x_continuous(breaks = seq(min(data_means$day), max(data_means$day), 1)) +
scale_color_manual(
  values = c(
    "25°C" = "#3B6FB6",
    "32°C" = "#D55E00"
  )
) +
scale_shape_manual(
  values = c(
    "Apo" = 16,
    "Inoc" = 17,
    "Sym" = 15
  )
) +
labs(color = "Temperature", shape = "Symbiotic state") +
theme(
  legend.position = c(0.60, 0.01),
  legend.justification = c("left", "bottom"),
  legend.background = element_blank()
)

```

p_sym_state_color_shape_2024

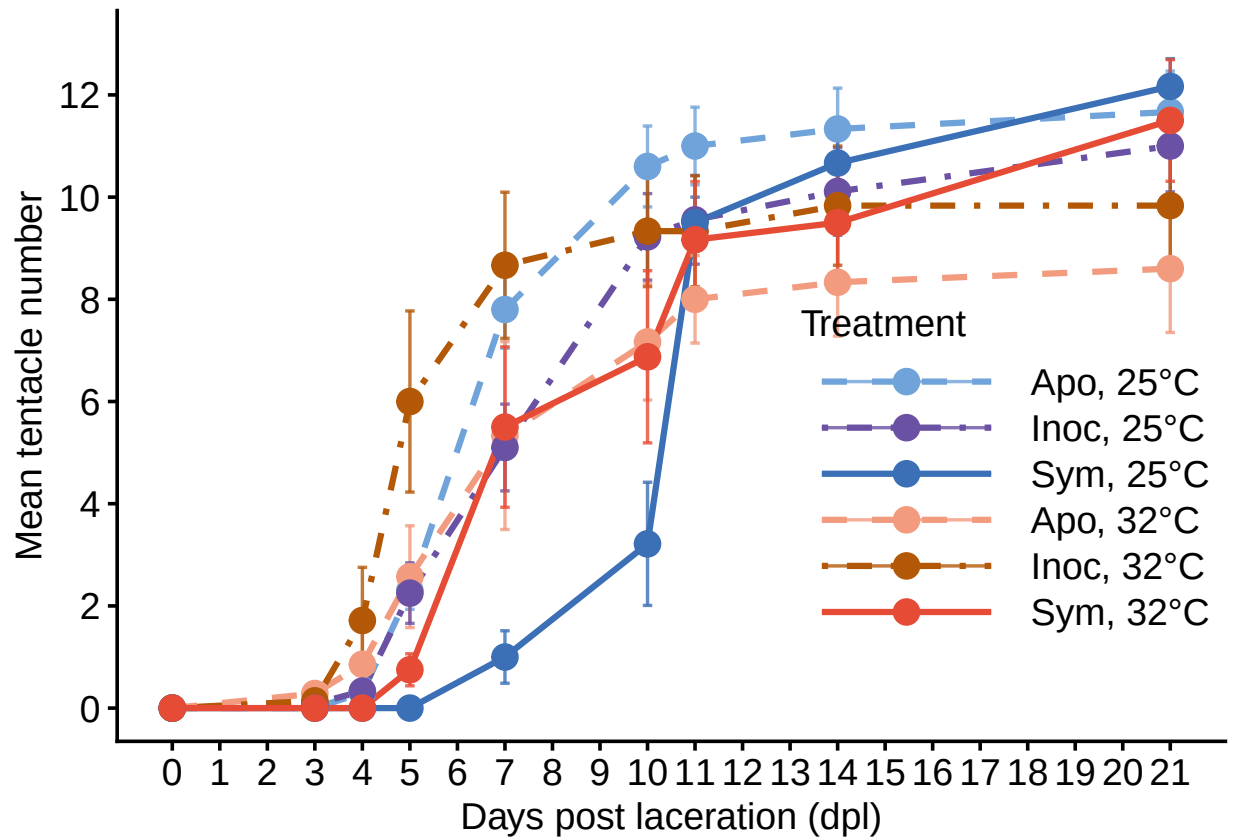


```
p_sym_state_color_linetype_2024 <- ggplot(
  data = data_means %>%
    filter(treatment %in% c(
      "H2-Apo-25", "H2-Ino-25", "H2-Sym-25",
      "H2-Apo-32", "H2-Ino-32", "H2-Sym-32"
    )) %>%
    mutate(
      treatment_label = case_when(
        treatment == "H2-Apo-25" ~ "Apo, 25°C",
        treatment == "H2-Ino-25" ~ "Inoc, 25°C",
        treatment == "H2-Sym-25" ~ "Sym, 25°C",
        treatment == "H2-Apo-32" ~ "Apo, 32°C",
        treatment == "H2-Ino-32" ~ "Inoc, 32°C",
        treatment == "H2-Sym-32" ~ "Sym, 32°C"
      ),
      treatment_label = factor(
        treatment_label,
        levels = c(
          "Apo, 25°C", "Inoc, 25°C", "Sym, 25°C",
          "Apo, 32°C", "Inoc, 32°C", "Sym, 32°C"
        )
      )
    )
),
```

```

aes(
  x = day,
  y = mean,
  color = treatment_label,
  linetype = treatment_label,
  group = treatment_label
)
) +
my_theme +
geom_line(linewidth = 1.1) +
geom_errorbar(
  aes(ymin = mean - se, ymax = mean + se),
  width = 0.2,
  linewidth = 0.6,
  alpha = 0.8,
  linetype = "solid"
) +
geom_point(size = 4) +
ylab("Mean tentacle number") +
xlab("Days post laceration (dpl)") +
scale_y_continuous(breaks = seq(0, 13, 2), limits = c(0, 13)) +
scale_x_continuous(breaks = seq(min(data_means$day), max(data_means$day), 1)) +
scale_color_manual(
  values = c(
    "Apo, 25°C" = "#6FA3D9",
    "Inoc, 25°C" = "#6A51A3",
    "Sym, 25°C" = "#3B6FB6",
    "Apo, 32°C" = "#F39B7F",
    "Inoc, 32°C" = "#B35806",
    "Sym, 32°C" = "#E64B35"
  )
) +
scale_linetype_manual(
  values = c(
    "Apo, 25°C" = "dashed",
    "Inoc, 25°C" = "dotdash",
    "Sym, 25°C" = "solid",
    "Apo, 32°C" = "dashed",
    "Inoc, 32°C" = "dotdash",
    "Sym, 32°C" = "solid"
  )
) +
labs(color = "Treatment", linetype = "Treatment") +
theme(
  legend.position = c(0.60, 0.12),
  legend.justification = c("left", "bottom"),
  legend.key.width = unit(2.8, "cm"),
  legend.background = element_blank()
)
p_sym_state_color_linetype_2024

```



```
p_sym_state_full_encoding_2024 <- ggplot(
  data = data_means %>%
    filter(treatment %in% c(
      "H2-Apo-25", "H2-Ino-25", "H2-Sym-25",
      "H2-Apo-32", "H2-Ino-32", "H2-Sym-32"
    )) %>%
    mutate(
      treatment_label = case_when(
        treatment == "H2-Apo-25" ~ "Apo, 25°C",
        treatment == "H2-Ino-25" ~ "Inoc, 25°C",
        treatment == "H2-Sym-25" ~ "Sym, 25°C",
        treatment == "H2-Apo-32" ~ "Apo, 32°C",
        treatment == "H2-Ino-32" ~ "Inoc, 32°C",
        treatment == "H2-Sym-32" ~ "Sym, 32°C"
      ),
      treatment_label = factor(
        treatment_label,
        levels = c(
          "Apo, 25°C", "Inoc, 25°C", "Sym, 25°C",
          "Apo, 32°C", "Inoc, 32°C", "Sym, 32°C"
        )
      )
    ),
  aes(
    x = day,
    y = mean,
```

```

    group = treatment_label,
    color = treatment_label,
    linetype = treatment_label,
    shape = treatment_label
  )
) +
my_theme +
geom_line(linewidth = 1.1) +
geom_errorbar(
  aes(ymin = mean - se, ymax = mean + se),
  width = 0.2,
  linewidth = 0.6,
  alpha = 0.8,
  linetype = "solid"
) +
geom_point(size = 4) +
ylab("Mean tentacle number") +
xlab("Days post laceration (dpl)") +
scale_y_continuous(
  breaks = seq(0, 13, 2),
  limits = c(0, 13)
) +
scale_x_continuous(
  breaks = seq(min(data_means$day), max(data_means$day), 1)
) +
scale_color_manual(
  values = c(
    "Apo, 25°C" = "#6FA3D9",
    "Inoc, 25°C" = "#6A51A3",
    "Sym, 25°C" = "#3B6FB6",
    "Apo, 32°C" = "#F39B7F",
    "Inoc, 32°C" = "#B35806",
    "Sym, 32°C" = "#E64B35"
  )
) +
scale_linetype_manual(
  values = c(
    "Apo, 25°C" = "dashed",
    "Inoc, 25°C" = "dotdash",
    "Sym, 25°C" = "solid",
    "Apo, 32°C" = "dashed",
    "Inoc, 32°C" = "dotdash",
    "Sym, 32°C" = "solid"
  )
) +
scale_shape_manual(
  values = c(
    "Apo, 25°C" = 16,
    "Inoc, 25°C" = 17,
    "Sym, 25°C" = 15,
    "Apo, 32°C" = 16,
    "Inoc, 32°C" = 17,
    "Sym, 32°C" = 15
  )
)

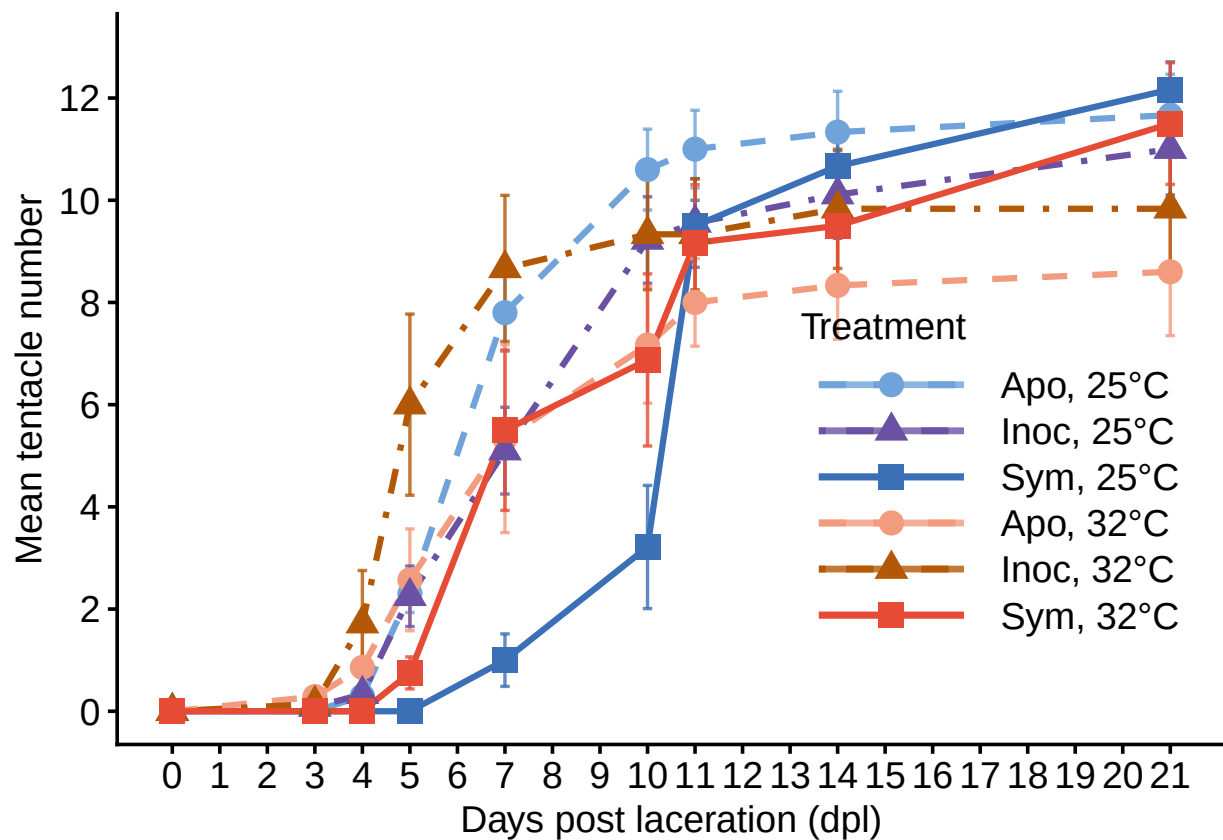
```

```

)
) +
labs(
  color = "Treatment",
  linetype = "Treatment",
  shape = "Treatment"
) +
guides(
  color = guide_legend(
    override.aes = list(
      linewidth = 1.2,
      size = 4
    )
  )
) +
theme(
  legend.position = c(0.60, 0.12),
  legend.justification = c("left", "bottom"),
  legend.background = element_blank(),
  legend.key.width = unit(2.4, "cm")
)

```

p_sym_state_full_encoding_2024



```

## Inoc only plot

data_means_inoc <- data_means %>%
  filter(treatment %in% c("H2-Inoc-25", "H2-Inoc-32")) %>%
  mutate(
    treatment_label = case_when(
      treatment == "H2-Inoc-25" ~ "Inoc, 25°C",
      treatment == "H2-Inoc-32" ~ "Inoc, 32°C"
    ),
    treatment_label = factor(
      treatment_label,
      levels = c("Inoc, 25°C", "Inoc, 32°C")
    )
  )

p_inoc_only <- ggplot(
  data_means_inoc,
  aes(
    x = day,
    y = mean,
    color = treatment_label,
    linetype = treatment_label,
    group = treatment_label
  )
) +
  my_theme +
  geom_line(linewidth = 1.2) +
  geom_errorbar(
    aes(ymin = mean - se, ymax = mean + se),
    width = 0.2,
    linewidth = 0.6,
    alpha = 0.8,
    linetype = "solid"
  ) +
  geom_point(size = 4) +

  annotate("text", x = 4, y = 3.5, label = "**", size = 6) +
  annotate("text", x = 5, y = 8.1, label = "***", size = 6) +
  annotate("text", x = 7, y = 10.8, label = "*", size = 6) +

  ylab("Mean tentacle number") +
  xlab("Days post laceration (dpl)") +

  scale_y_continuous(
    breaks = seq(0, 12, 2),
    limits = c(0, 12)
  ) +
  scale_x_continuous(
    breaks = seq(min(data_means_inoc$day), max(data_means_inoc$day), 1)
  ) +
  scale_color_manual(
    values = c(
      "Inoc, 25°C" = "#3B88C3",

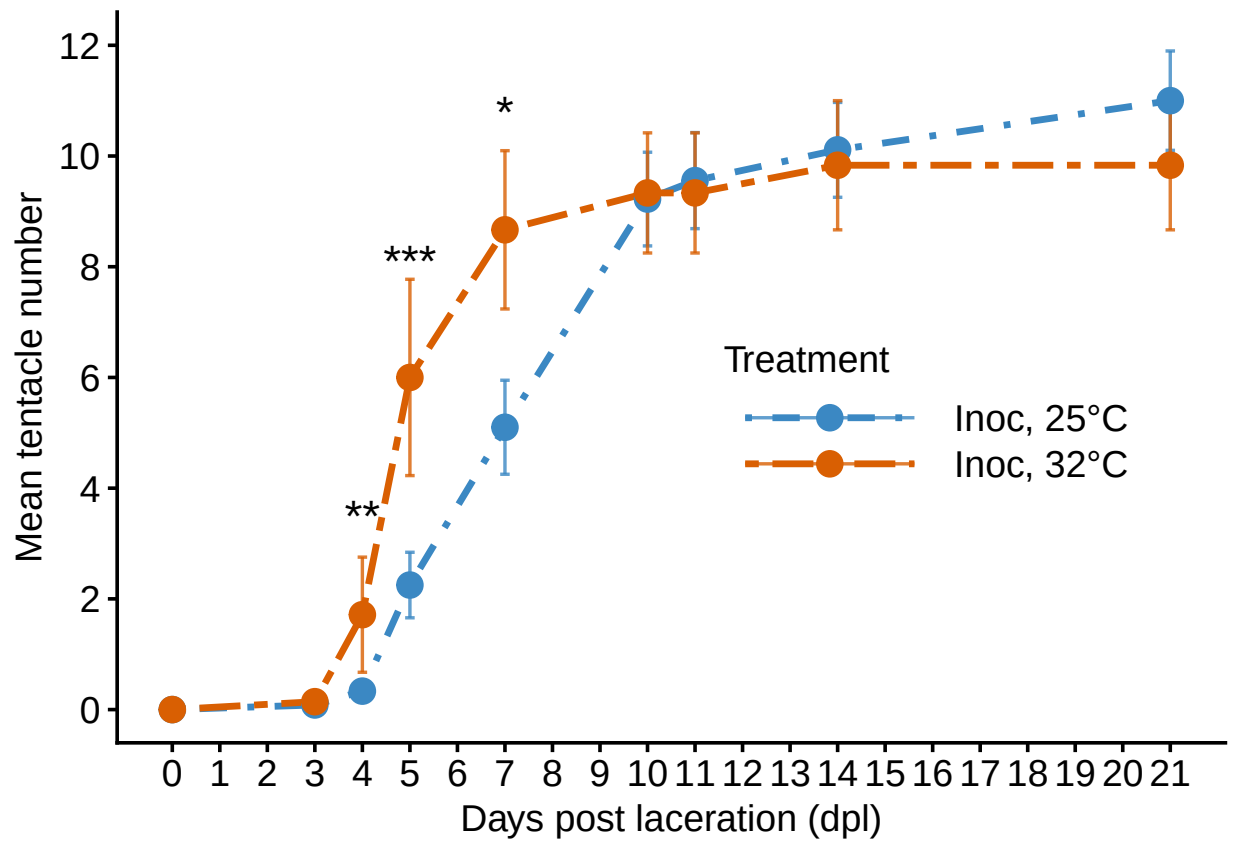
```

```

    "Inoc, 32°C" = "#D95F02"
  )
) +
scale_linetype_manual(
  values = c(
    "Inoc, 25°C" = "dotdash",
    "Inoc, 32°C" = "twodash"
  )
) +
labs(
  color = "Treatment",
  linetype = "Treatment"
) +
theme(
  legend.position = c(0.73, 0.45),
  legend.justification = c("center", "center"),
  legend.key.width = unit(2.8, "cm")
)

```

p_inoc_only



```

## Symbiotic state stats

df_sym_state <- ursas2024_data %>%
  filter(

```

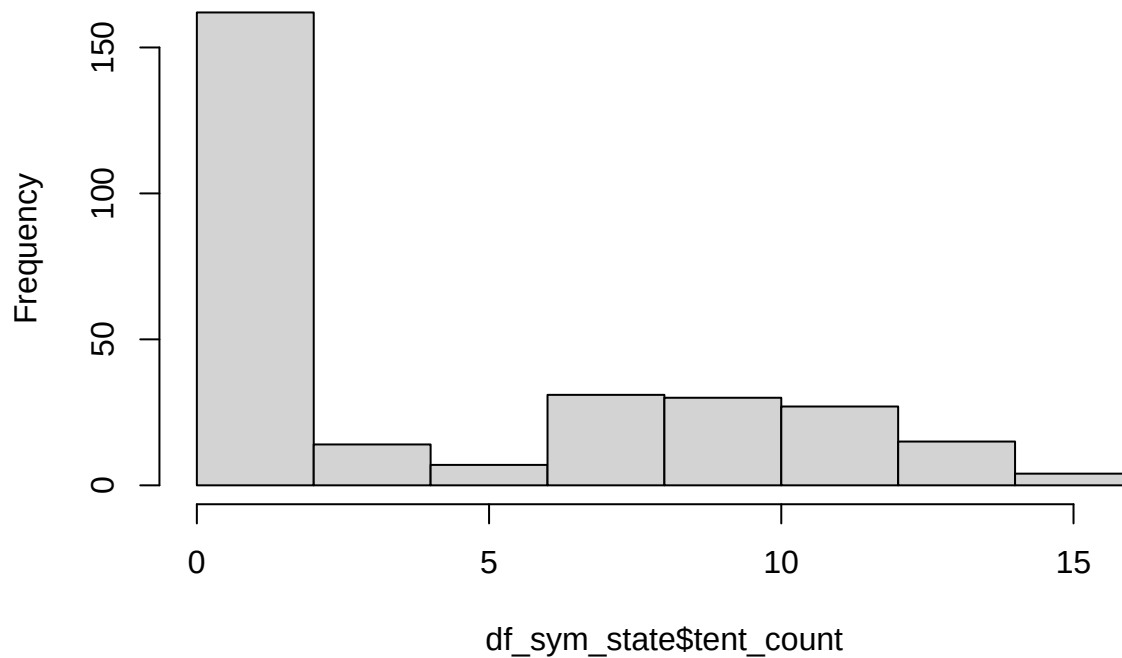
```

line == "H2",
treatment %in% c("H2-Apo-25", "H2-Ino-25", "H2-Sym-25")
) %>%
mutate(
  treatment = factor(treatment, levels = c("H2-Apo-25", "H2-Ino-25", "H2-Sym-25")),
  day = factor(day),
  id = factor(id)
)
df_sym_state$treatment <- factor(df_sym_state$treatment)
df_sym_state$day <- factor(df_sym_state$day)
df_sym_state$id <- factor(df_sym_state$id)

hist(df_sym_state$tent_count)

```

Histogram of df_sym_state\$tent_count

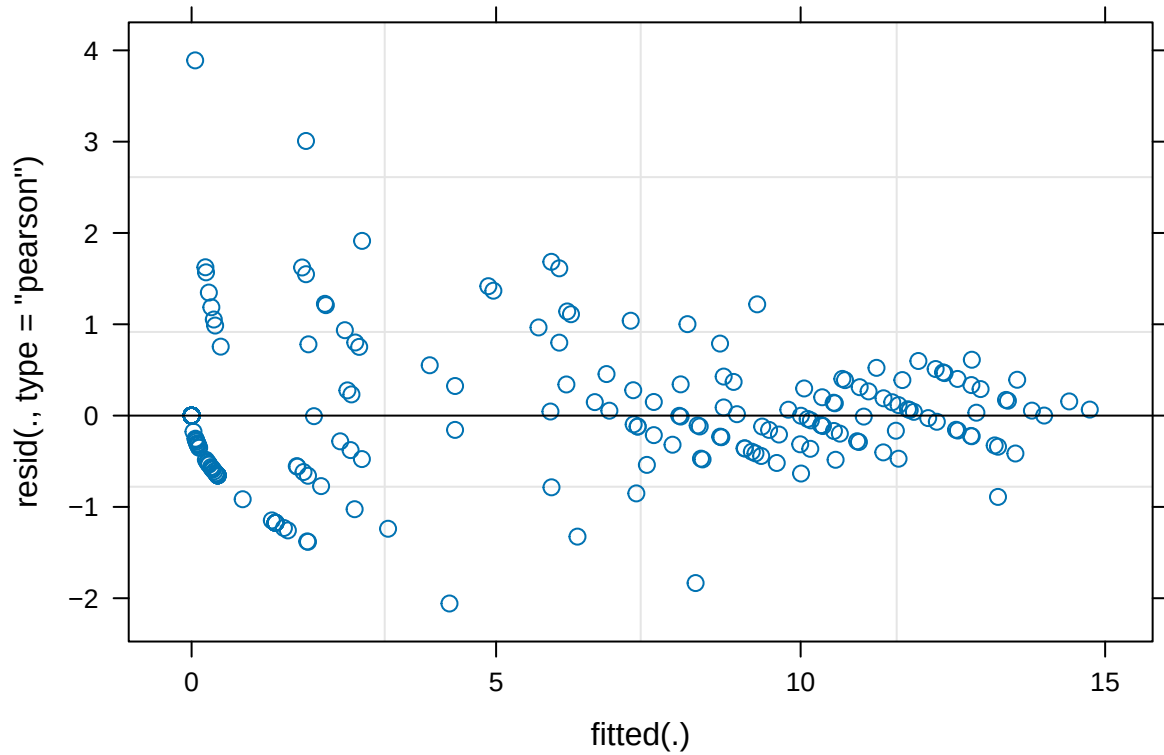


```

model_sym_state <- glmer(
  tent_count ~ treatment * day + (1 | id),
  family = poisson(link = "log"),
  data = df_sym_state,
  control = glmerControl(
    optimizer = "bobyqa",
    optCtrl = list(maxfun = 200000)
  )
)

plot(model_sym_state)

```

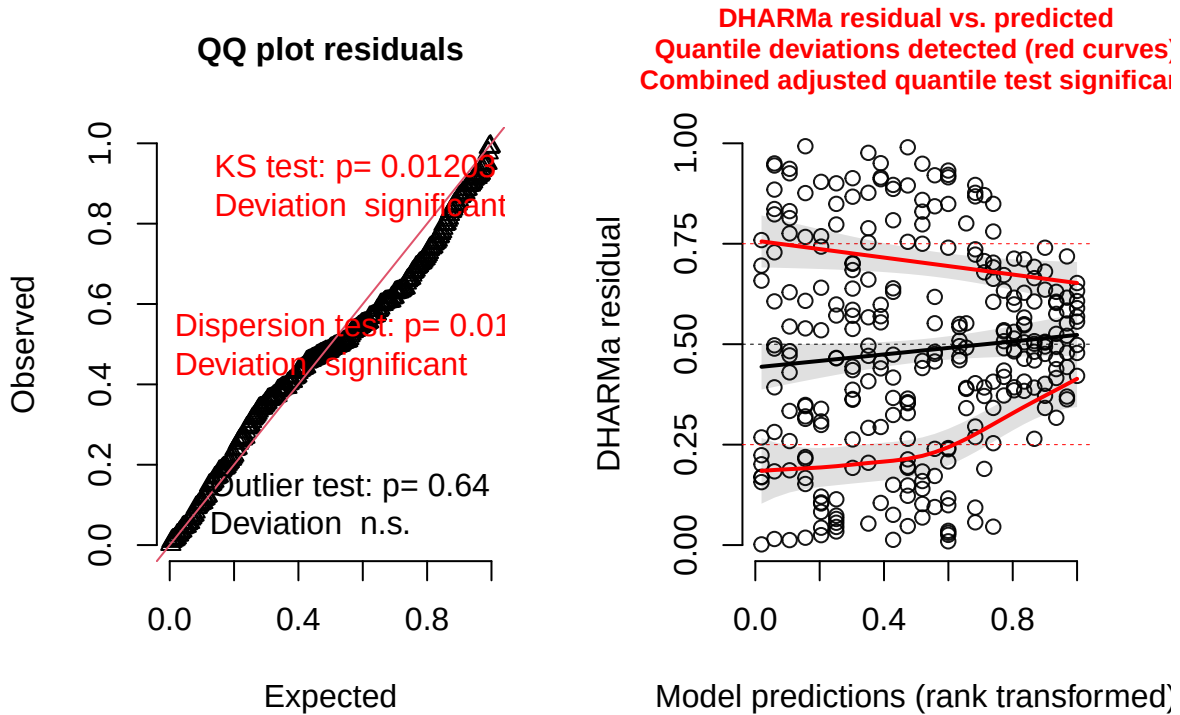


```
check_overdispersion(model_sym_state)
```

```
## # Overdispersion test (using simulated residuals)
##
## dispersion ratio = 0.691
## p-value = 0.016
```

```
sim_res_sym <- simulateResiduals(model_sym_state)
plot(sim_res_sym)
```

DHARMa residual



```
Anova(model_sym_state, type = "II")
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: tent_count
##           Chisq Df Pr(>Chisq)
## treatment    50.977 12  9.401e-07 ***
## day          264.719 18 < 2.2e-16 ***
## treatment:day  27.903 16  0.03247 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emm_sym <- emmeans(model_sym_state, ~ treatment | day)
```

```
pairs(emm_sym, adjust = "tukey")
```

```
## day = 0:
## contrast          estimate      SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) -3.22e-02 1.62e+04 Inf  0.000  1.0000
## (H2-Apo-25) - (H2-Sym-25) -3.31e-02 1.49e+04 Inf  0.000  1.0000
## (H2-Ino-25) - (H2-Sym-25) -9.56e-04 1.46e+04 Inf  0.000  1.0000
##
## day = 3:
## contrast          estimate      SE df z.ratio p.value
```

```

## (H2-Apo-25) - (H2-Ino-25) -1.88e+01 1.33e+04 Inf -0.001 1.0000
## (H2-Apo-25) - (H2-Sym-25) -2.49e-02 1.65e+04 Inf 0.000 1.0000
## (H2-Ino-25) - (H2-Sym-25) 1.87e+01 9.83e+03 Inf 0.002 1.0000
##
## day = 4:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) -4.87e-02 8.03e-01 Inf -0.061 0.9980
## (H2-Apo-25) - (H2-Sym-25) 2.01e+01 9.82e+03 Inf 0.002 1.0000
## (H2-Ino-25) - (H2-Sym-25) 2.01e+01 9.82e+03 Inf 0.002 1.0000
##
## day = 5:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 7.86e-02 3.73e-01 Inf 0.211 0.9758
## (H2-Apo-25) - (H2-Sym-25) 2.21e+01 9.80e+03 Inf 0.002 1.0000
## (H2-Ino-25) - (H2-Sym-25) 2.20e+01 9.80e+03 Inf 0.002 1.0000
##
## day = 7:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 4.83e-01 3.06e-01 Inf 1.577 0.2555
## (H2-Apo-25) - (H2-Sym-25) 2.31e+00 3.80e-01 Inf 6.091 <0.0001
## (H2-Ino-25) - (H2-Sym-25) 1.83e+00 3.86e-01 Inf 4.736 <0.0001
##
## day = 10:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 2.63e-01 2.89e-01 Inf 0.909 0.6348
## (H2-Apo-25) - (H2-Sym-25) 1.45e+00 3.01e-01 Inf 4.819 <0.0001
## (H2-Ino-25) - (H2-Sym-25) 1.19e+00 3.04e-01 Inf 3.904 0.0003
##
## day = 11:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 2.65e-01 2.88e-01 Inf 0.918 0.6287
## (H2-Apo-25) - (H2-Sym-25) 9.51e-01 3.09e-01 Inf 3.075 0.0060
## (H2-Ino-25) - (H2-Sym-25) 6.86e-01 3.12e-01 Inf 2.198 0.0716
##
## day = 14:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 2.54e-01 2.89e-01 Inf 0.879 0.6535
## (H2-Apo-25) - (H2-Sym-25) 8.80e-01 3.08e-01 Inf 2.862 0.0117
## (H2-Ino-25) - (H2-Sym-25) 6.27e-01 3.08e-01 Inf 2.035 0.1040
##
## day = 21:
## contrast estimate SE df z.ratio p.value
## (H2-Apo-25) - (H2-Ino-25) 1.98e-01 2.87e-01 Inf 0.692 0.7679
## (H2-Apo-25) - (H2-Sym-25) 7.78e-01 3.04e-01 Inf 2.559 0.0283
## (H2-Ino-25) - (H2-Sym-25) 5.79e-01 3.03e-01 Inf 1.909 0.1360
##
## Results are given on the log (not the response) scale.
## P value adjustment: tukey method for comparing a family of 3 estimates

```